VoltGuard weekly summary · region CEN

Report VG-WK-CEN-2026-W44 · generated 28 Oct 2026, 10:13 MYT · for revenue-protection managers, region CEN

Meter data: 24–30 Sep 2026
Field work: 21–28 Oct 2026

SAMPLE LAYOUT. Figures taken from the current report are real; figures marked † are illustrative, to show what the generated report should contain. Simulated data (R0): not TNB data.

This week in one paragraph. Unexplained energy in region CEN averaged **285 kWh a day**, 2.5% lower than the week before, so the leak is steady, not growing. **Two of three zones remain flagged**; TX-CEN-002 alone accounts for 55% and draws a flat 24/7 load typical of mining rigs. No cases were opened or closed this week and no inspections were recorded.

- **Action 1:** run the case job for TX-CEN-002 and TX-CEN-001; their top suspects have no open case.
- **Action 2:** add random routine checks: 4 so far is too few to measure the model's real hit rate.

1 · Key figures

2 of 3
Flagged zones
Unexplained energy above the flag rule for 7 days in a row
no change vs last week†

285
Unexplained kWh per day
Average over the meter week, all zones with a transformer meter
−2.5% vs 292 the week before

~8,550
kWh a month at this level
285 × 30 days. Value in RM: [TNB tariff: to be supplied]
estimate, see section 3

0 / 0
Cases opened / closed
During the field-work week
10 closed in total

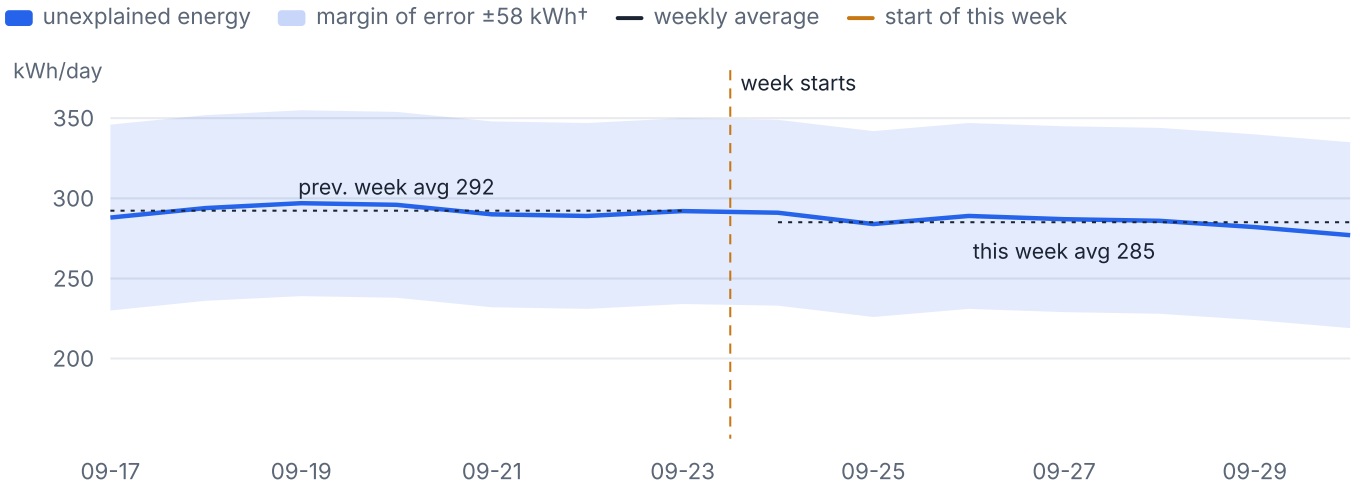
0
Overdue cases
Assigned and not visited within 72 h

0 · 0
Reports exported · open alerts
Reports after both approvals · data or model alerts

Why two dates? Meter data reach VoltGuard daily, so energy figures cover the latest full week of readings (24–30 Sep). Cases and inspections are counted over the last 7 calendar days (21–28 Oct). In this prototype the simulated meter data end on 30 Sep, which is why the two windows are a month apart; with live TNB data they would be one or two days apart.

2 · Is energy still going missing?

Unexplained energy per day, whole region, last 14 days

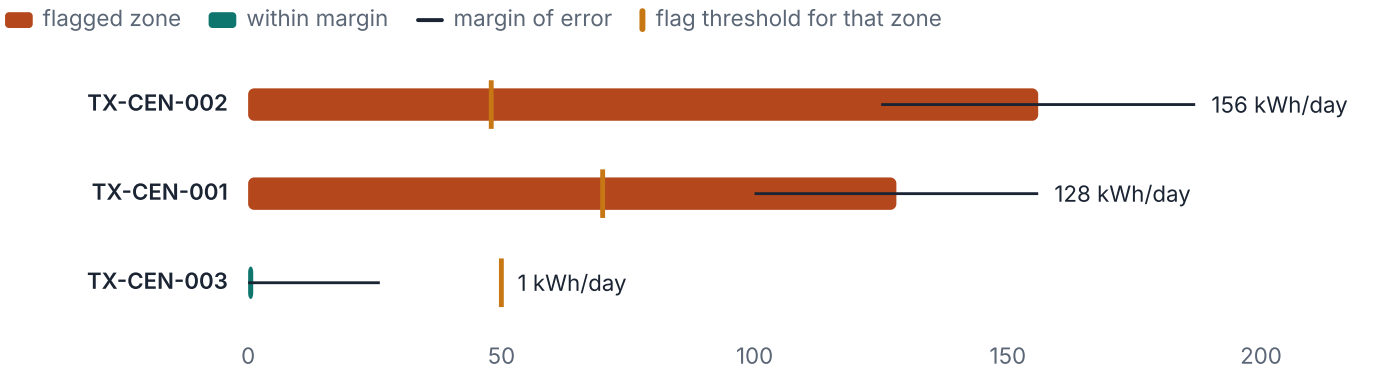


What this shows: for each day, energy the transformers supplied minus what customers' meters recorded, minus the estimate for monthly-read meters and the expected technical loss in cables and transformers. **How to read it:** a line inside the shaded band could be measurement error; a line well above it means energy is leaving the network unrecorded.

Takeaway: about 285 kWh a day goes unrecorded, well above the margin of error and steady for two weeks. A steady, flat leak points to a continuous load, not a one-off.

3 · Where is it going missing?

Unexplained energy per zone this week



How to read it: a zone is flagged when its bar, minus the margin of error, stays past its threshold mark for 7 days. Thresholds differ by zone because they scale with the energy each transformer supplies (5% of supply, or a minimum amount per day, whichever is larger).

Zone	District	kWh/day	% of supply	Margin	Threshold	Status	Leak started	Daily pattern	Priority	Open cases
TX-CEN-002	District A†	156	16.2%†	±31†	48†	Flagged	2026-09-08†	flat 24/7 (CV 0.12)†	Large load†	0
TX-CEN-001	District B†	128	9.1%†	±28†	70†	Flagged	since data start†	varies by hour†	Normal†	0
TX-CEN-003	District A†	1	0.1%†	±25†	50†	Within margin	–†	–†	–†	0

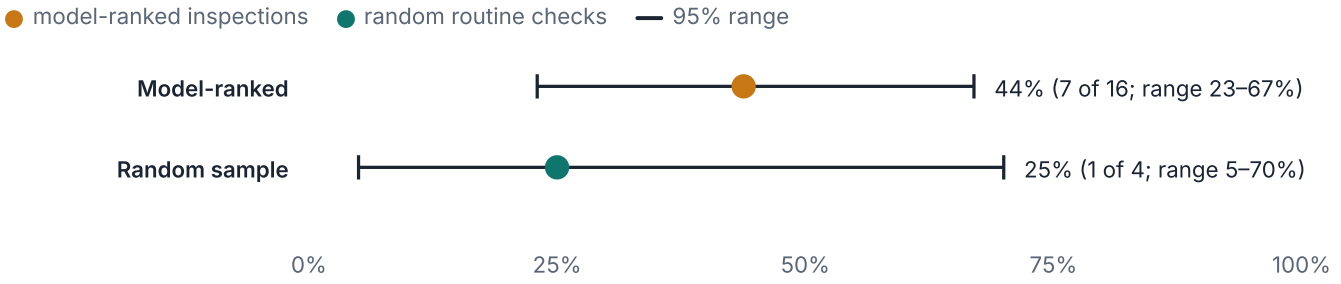
Takeaway: TX-CEN-002 is the priority: the largest leak, above 15% of its supply, flat around the clock. TX-CEN-003's 1 kWh a day sits inside its margin of error, so it is not flagged.

Value at risk (estimate)

285 kWh a day × 30 days ≈ **8,550 kWh a month**. In RM: 8,550 × [TNB tariff per kWh: to be supplied]. This counts all unexplained energy, including the margin of error, so it is an upper estimate; counting only energy above each zone's threshold gives about **5,000 kWh a month**†.

4 · Are inspections finding theft?

Hit rate so far, with 95% ranges



Inspections	Labelled	Confirmed theft	Hit rate	95% range	What it means
Model-ranked	16	7	43.8%	23–67%	Homes chosen by VoltGuard's ranking
Random sample	4	1	25.0%	5–70%	Homes picked at random in flagged zones: the fair baseline

How to read it: the hit rate is the share of inspected homes where theft was confirmed. Random routine checks show what you would find without the model, so the gap between the two rows is the model's value. The 95% range shows how much the rate could move by chance with so few inspections.

Takeaway: model-ranked inspections look better (44% vs 25%), but the ranges overlap heavily: four random checks are too few to prove a difference. No inspections were recorded this week. Outcomes so far come from simulated cases, so they show how the measurement works, not real performance.

Inspection outcomes this week

Outcome	This week	All time†
Confirmed theft	0	8
False alarm	0	10
Inconclusive	0	2
No access (not used for learning)	0	1

Cases by status now

Status	Cases	Meaning
New	0	Waiting for supervisor
Assigned	0	Team sent
Visited	0	Outcome recorded
Reporting	0	Theft confirmed, report in approval
Closed	10	Finished

5 · Can these figures be trusted?

Data quality (meter week)

Meter-days with ≥ 90% of readings	98.7%†
Meters read monthly (estimated daily)	12%†
Zones without a transformer meter	0
Duplicate or out-of-order readings fixed	14†
Fallbacks in use	Voltage is simulated (AS-3)

Model in use

Scoring version	R0-2026-10-27-a†
Last retrained	27 Oct 2026, on 20 labels†
Data or drift alerts this week	0
Thresholds	5% of supply or 15 kWh/day, 7 days; large load above 15%

Assumptions and limits

- **Simulated data.** This prototype runs on a simulated region built from public household profiles; voltages come from a network model, because open data hold energy readings only.
- **Technical loss is estimated** by an engineering model of cables and transformers; its error is part of the margin of error.
- **Monthly-read meters** are spread across the month using similar homes' daily patterns; this also widens the margin.
- **Small leaks can hide:** a tap of about 1% of a zone's supply sits inside the margin and will not flag the zone.
- **Scores rank risk;** they are not findings of theft. People decide after inspection.
- **Tariff not yet supplied,** so value at risk is shown in kWh only.
- **Zone-level only:** this report holds no premise IDs, names or addresses, and covers region CEN only.

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6 · Recommended next steps

1. Supervisor: run the case job for TX-CEN-002 and TX-CEN-001; suggest a joint operation with police for TX-CEN-002 (flat, large load).
2. Manager: approve at least 6 more random routine checks this month so the hit-rate comparison becomes meaningful†.
3. Data team: confirm the TNB tariff for the value-at-risk figure.

Steps are generated from fixed rules on the figures above, not written by the language model.

Glossary

Unexplained energy	Supplied by the transformer minus metered use, minus estimates for monthly-read meters and expected technical loss.
Margin of error	How far unexplained energy can stray by chance, from meter accuracy, monthly-read estimates and the loss model.
Flagged zone	Unexplained energy above the threshold plus the margin of error for 7 days in a row.
Flat 24/7	The leak hardly changes across the day (coefficient of variation below 0.2), typical of mining rigs.
Hit rate	Share of inspected homes where theft was confirmed.
Random routine check	An inspection of a home picked at random in a flagged zone, used as the fair baseline.